MILITARY SPECIFICATION

TEST SET AN/AWM-80 for the

AN/AWG-10A Missile Control System

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1.0 SCOPE

1.1 Scope

This specification covers the AN/AWM-80 Missile Control System Test Set to be used at the Intermediate Level of Maintenance to maintain the AN/AWG-10A Computer consisting of: LRU-15 and LRU-16; the Test Control-Indicator, LRU-14; and the Servoed Optical Sight, part of the Command Indicator, LRU-9.

1.2 Classification

The complete AN/AWM-80 MCS Test Set shall be comprised of the Test Set Group, OQ-230/AWM-80 and the Test Set, Optical Sight AN/AWM-81.

1.2.1 The following items describe the Test Set Group, OQ-230/AWM-80.

ITEM	NOMENCLATURE	REF PARA.
Control, Power Supply	C-10015/AWM-80	3.7.1.5.1
Reader, Punched Tape	RP-236/AWM-80	3.7.1.5.2
Transport, Punched Tape	MX-9782/AWM-80	3.7.1.5.3
Computer, Test	CP-1263 (P) AWM	3.7.1.5.4
Maintenance Panel		3.7.1.5.5
Converter-Interface Unit	CV-3347/AWM-80	3.7.1.5.6
Blower Assembly		3.7.1.5.7
Power Supply Assembly	PP-7284/AWM-80	3.7.1.5.8
Connector Plate Assembly		3.7.1.5.9
Fixture Cooler, Air	MT-4813/AWM-80	3.7.1.5.10
Interconnecting Box	J-3388/AWM	3.7.1.5.11
Interlock Assembly		3.7.1.5.12
Top Plate Assembly		3.7.1.5.13

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 (Continued)

ITEM	NOMENCLATURE	REF PARA
Isolation Transformer		3.7.1.5.14
Cabinet, Electrical Equipment	CY-7532/AWM-80	3.7.1.5.15

1.2.2 The following items describe the Test Set, Optical Sight, AN/AWM-81.

ITEM	NOMENCLATURE	REF PARA
Test Set, Optical Sight	TS-3496/AWM-81	3.7.2.2.1
Test Set, Holding Fixture	MT-4817/AWM-81	3.7.2.2.2

2.0 APPLICABLE DOCUMENTS

2.1 General

The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-27D	Transformers and Inductors (Audio Power and High Power Pulse), General Specification for
MIL-E-15090	Enamel, Equipment Light Gray (Formula No. 111)
MIL-F-15733/2	Filters, Radio Interference Style FL24
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-5422	Environmental, Airborne, Electronic and Associated Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-28800B	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

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2.1 (Continued)

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential distribution
MIL-STD-794	Parts and Equipment, Procedures for Packing and Packaging of

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems; General Requirement for
AR-30A	Integrated Logistic Support Program Requirements for Aeronautical Systems and Equipment

WESTINGHOUSE DRAWINGS

90C99015	Interlock Assembly

2.1.1 Availability of documents

When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3.0 . REQUIREMENTS

3.1 Testing - This specification makes provision for testing of the AN/AWM-80 Test Set (See 4).

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3.0 (Continued)

3.2 Materials, Parts, and Processes - In the selection of parts, materials, and processes, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

a. Parts, Materials, and Processes selected according to the requirements of MIL-T-28800 shall be given first consideration.

b. Guidelines for the selection of nonstandard parts shall be per MIL-T-28800, Selection of Parts and Materials.

c. In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass electrical environmental test.

d. Paint for consoles for test equipment used at shop level maintenance shall be baked enamel per MIL-E-15090 except that the color shall be light gray, formula #111.

3.2.1 Nonstandard parts

Nonstandard parts and materials shall be as outlined in MIL-T-28800, Materials, Parts and Processes, except that mechanical parts shall be considered as nonstandard parts only when the material requirements of MIL-T-28800, as modified herein, are not met.

3.2.2 Semiconductor Types

The choice and application of semiconductor types used in the equipment shall be in accordance with MIL-T-28800 and AR-5.

3.2.3 Noncorrosive Hardware

External Military Standard (MS) hardware which is used during servicing or installing test equipments shall be of corrosive-resistant type and shall not be further finish-treated.

3.3 Design and Construction

Unless otherwise specified herein or in the detail equipment specification, Contractor-Furnished Equipment (CFE) shall conform with all the requirements of MIL-T-28800 as applicable for design, construction, and workmanship. Government Furnished Equipment (GFE) shall conform to the applicable detail equipment specification.

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3.3 (Continued)

3.3.1 Total weight - The total weight of the AN/AWM-80 Test Set, including cables, shall be a minimum consistent with good design and shall not exceed 900 pounds (excluding the AN/AWM-81 Test Set).

3.3.2 Microelectronic items - When used, microelectronic items shall be in accordance with AR-5.

3.3.3 Modules, maintenance - The electronic portions of the equipment shall be divided into maintenance modules as specified in AR-10. Modules shall normally be considered repairable.

3.3.4 Cables and Connectors - The test set shall provide for the use of cables and connectors in accordance with MIL-T-28800. Insofar as is practical, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.4.1 Cabling - The test set cabling shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested except that spare wires shall be contained in the cables for future expansion or repairs, as required.

3.3.4.2 External Connectors - External connectors shall provide for the following:

Reference Designation	Connector Type	Function
A8J1	M81511/46FF01P1	Test Interface to 16J2
A8J2	M81511/41FH01S2	Computer I/O to 15J3/16J1
A8J3	M81511/41FH01S1	Computer I/O to 15J4
A8J4	M81511/41FH02P1	Computer I/O to 15J1/ 16J2

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3.3.4.2 (Continued)

Reference Designation	Connector Type	Function
A8J5	M81511/41FD02P1	Computer Power In to 16J3
A8J6	MS3126F24-61S	LRU 14 I/O to 14J1
A8J7	MS3126F24-61P	LRU 14 I/O to 14J2
9A5J1	97942-577R854H06	INPUT/OUTPUT - SOS
9A5J2	M24308/2-283	INPUT/OUTPUT - SOS

3.3.5 Interchangeability - Unless otherwise specified herein or in the detail equipment specification, the equipment shall meet the interchangeability requirements as defined in MIL-T-28800.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-28800. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance. A maintenance panel shall be incorporated into the test set to provide the capability of controlling the internal test set processor along with the capability for manual entry of instructions and display of the various registers and memory contents. A self test capability shall be included to determine the functional status of the test set. The self-test program shall be contained on a separate punched mylar tape for transfer into the test set memory and shall provide a GO/NO-GO indication of the operational status of the test set.

3.3.7 Human Engineering - Human engineering shall be in accordance with MIL-T-28800.

3.4 Availability - The availability shall be a minimum of 99 percent when calculated using the results of the reliability and maintainability demonstrations of paragraphs 3.4.1 and 3.4.2 as follows:

$$\text{Availability} = \frac{\theta}{\theta + \bar{M}_{ct}}$$

where:

θ is the observed Mean-Time-Between-Failure (MTBF) equal to the total operating time of the equipment divided by the number of failures, as defined in MIL-STD-781.

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3.4 (Continued)

and:

$\bar{M}_{ct}$ is equal to the mean corrective maintenance downtime, defined in MIL-STD-471.

3.4.1 Reliability - The contractor shall conduct a reliability program in accordance with MIL-T-28800. The program plan shall be approved by the procuring activity.

3.4.1.1 Operating Stability - The equipment shall operate as specified for not less than 800 hours or 12 months, whichever comes first, without scheduled maintenance, realignment, or calibration.

3.4.1.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours. Parts requiring scheduled replacement due to wear during the life of the equipment and the wearout life of such parts shall be determined by the contractor and submitted to the procuring activity.

3.4.1.3 Reliability in Mean-Time-Between-Failure (MTBF) - The equipment, including any built-in-test provisions, shall have a minimum acceptable (operating) MTBF, (θ_1) of 400 hours when tested as specified in 4.4.3. The equipment shall have an inherent design reliability of 812 hours, (nominal) MTBF.

3.4.2 Maintainability - The contractor shall conduct a maintainability program in accordance with MIL-T-28800. The program plan shall be approved by the procuring activity. Built-in-test features, modular construction and packaging, and provisions for test points shall be as specified in AR-10. The program will provide appropriate input data for Integrated Logistic Support requirements in accordance with AR-5 or AR-30 applicable to the avionic equipment to be tested by this support equipment.

3.4.2.1 Maintenance Times - The equipment, including any built-in provisions, shall be designed to meet the following quantitative maintainability requirements for corrective maintenance when tested in accordance with 4.2.2.1.

a. Mean time to repair, $\bar{M}_{ct}$ 2.5 hours.

b. Maximum time to repair, M_{max} 4.4. hours.
(at the 90th percentile).

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3.4.2.2 Maintenance manhours per maintenance action - The total maintenance action shall not exceed 5 hours when conducted in accordance with 4.2.2.1.

3.4.3 Probability of successful performance of specified test (Compatability) - The probability of successful performance during a complete test (all test sequences) shall be a minimum of 98.0 percent when tested in accordance with provisions of 4.2.2.2. The probability of successful performance of a complete test of the avionic equipment to be tested by this support equipment shall include:

a. not detecting a failure in the avionic equipment to be tested,

b. indicating a non-existent failure in the avionic equipment to be tested,

c. the probability of causing a failure in the avionic equipment to be tested, and

d. The probability that a failure in the avionic equipment under test will cause a failure in the support equipment (except for the activation of a protective device).

3.4.4 Interference Control - The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be controlled within the limits of MIL-STD-461 unless otherwise specified.

3.4.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.6. Service Conditions - Service conditions (environmental) shall be in accordance with MIL-T-28800.

3.4.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90 percent relative humidity
Input Voltage	3-phase, 115 $\pm$ 11.5 Volts (V)

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3.4.7 (Continued)

line to neutral, 400 Hz ± 20 Hz. 27 ± 2 Volts Direct Current (DC)

3.5 Transportability - Transportability requirements shall be in accordance with MIL-T-28800.

3.6 Performance Characteristics

3.6.1 General

Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein, or in the detail specification of the equipment being tested.

3.6.2 Functions

The MCS Test Set is required to verify the overall functions of the computer, consisting of Line Replaceable Unit - 15 (LRU) and 16; the Control - Indicator Test, LRU-14; and the servoed optical sight, part of LRU-9, to detect any condition which results in a failure and to rapidly and simply isolate the cause of a failure to one of the plug-in Shop Replaceable Assemblies (SRA) within the unit under test (UUT).

3.6.3 Form Factor

A 1 1/2 bay electrical equipment cabinet provides the facility for mounting the sub-assemblies of the test set, for housing the interconnecting cabling and the AN/AWM-81 Test Set, and for input of the prime power from the ship's supply.

3.6.4 Frequency Characteristics

3.6.4.1 Range

DC

400 Hz (AC)

Digital data at 31.25 KHz clock rate

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3.6.4.2 Accuracy

All input and output voltages shall be controlled in frequency to within ± 5%.

3.6.4.3 Operational Stability

Unless otherwise specified in the individual equipment specification, the equipment shall operate with satisfactory performance, continuously or intermittently, for a period of 24 hours without necessity for readjustment of any operator controls which are inaccessible to the operator during use, except where adjustments associated with parts failures are required following replacement of those parts.

3.6.5 Modulation Characteristics

None

3.6.6 Indication

Visual indications shall include lamp monitors and GO/NO-GO indicators. Numeric displays in the form of Light Emitting Diode readouts shall provide instructions and intelligence to the operator as required by the particular test being performed.

3.6.7 Output Voltages

The test set shall provide outputs to the Unit-Under-Test as required to fully exercise that unit. The following voltages are required as inputs to the UUT.

a. LRU 15/16

Discretes 0 or 28 VDC
DC Analog Neg 10 VDC to Pos 43 VDC
AC Analog (400 Hz) 0 to 24 V (rms) at phase angles of 0° or 180°
(relative to phase A).

b. LRU 14

Discretes 0 or 28 VDC
DC Signal Neg 100 VDC to Pos 250 VDC
AC Signal (400 Hz) 2.5V and 5.0V (rms)
Digital NRZ data level at 31.25 KHz rate.

c. LRU 9A5

Discretes 0 or 28 VDC

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3.6.7 (Continued)

DC Signals Neg 12 VDC to Pos 12 VDC (Variable)
AC (400 Hz) 0 or 5 V (rms) selected

24 V (rms)

3.6.8 Electrical Power Source - The test equipment shall operate from a power source as specified below:

AC 3-phase, 115 ± 11.5 V (rms)
line to neutral, 380 to 420 Hz,
5.2 KW (max)

DC 27 ± 2 VDC,
150 watts (max)

3.6.8.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704. Circuit breakers shall be used to protect the test set from short circuits or overloads in the Unit-Under-Test.

3.6.9 Warm-up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

3.7 Detailed Requirements

3.7.1 Test Set Group, MCS OQ-230/AWM-80

3.7.1.1 Function - The MCS Test Set Group shall be for shop use and shall monitor outputs, provide stimuli, mounting and cooling air for LRUs 14, 15 and 16 of the MCS. It shall also provide power for operation of the MCS servoed optical sight (SOS). The equipment shall electrically simulate required aircraft flight functions for the above units and shall contain provisions for fault isolation, adjustment and replacement of defective LRU Shop Replaceable Assemblies (SRAs).

3.7.1.2 Form Factor - The test set shall be housed in a 1 1/2 bay console containing the items specified herein. The maximum dimensions of this unit shall be 24" (wide) by 25.5" (deep) by 72" (high) for bay 1 and 29" (wide) by 25.5" (deep) by 36" (high) for bay 2. The weight of the MCS Test Set Group shall not exceed 900 punds, including the interconnecting cabling, shock mounts and bottom frame.

3.7.1.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-28800, Class 3.

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3.7.1.4 Primary Input Power Requirements - The equipment shall require power of the following type and shall not exceed the specified amounts.

AC Power: 3-phase, 115/208 ± 10 percent, 380 to 420 Hz, 15 Amperes (A)/phase

DC Power: 27 ± 2 VDC, 10 Amperes

3.7.1.5 Contents of Unit - The MCS Test Set Group shall contain the following assemblies:

a. Control, Power Supply

b. Reader, Punched Tape

c. Transport, Tape

d. Computer, Test

e. Maintenance Panel (P.O. Computer, Test)

f. Converter - Interface Unit

g. Blower Assembly

h. Power Supply Assembly

i. Connector Plate Assembly

j. Fixture Cooler, Air

k. Interconnecting Box

l. Interlock Assembly

m. Top Plate Assembly

n. Isolation Transformer

o. Cabinet Electrical Equipment

3.7.1.5.1 Control Power Supply

3.7.1.5.1.1 Functions - The Power Control Panel shall provide the following functions:

a. Monitor the three phase voltage input and provide a phase detection interlock to prevent operation of the test set in case of a phase reversal or loss of one or more phases of prime power.

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3.7.1.5.1.1 (Continued)

b. Overcurrent protection for each phase of input voltage (e.g. circuit breaker).

c. Distribute 3-phase AC power and 28 VDC power to all units within the test set which require them.

d. Develop 10 VRMS synchro reference voltage.

3.7.1.5.1.2 Form Factor - This unit shall not exceed 13 by 19 by 3.5 inches. The weight of the unit shall not exceed 6 pounds.

3.7.1.5.1.3 Indicators - The following monitors and indicating lights shall be provided on the front panel.

a. 28 VDC - The 28 VDC indicator shall be an incandescent indicator light with a translucent white lens which shall show when the 28 VDC line voltage is connected and all interlocks are closed.

b. 115V 400 Hz - The 115V 400 Hz indicator shall be an incandescent indicator light with a translucent white lens which shall show when the three phase 115V 400 Hz line voltage is connected in the proper sequence and all interlocks are closed.

3.7.1.5.1.4 Protective Devices - The following protective devices shall be provided.

a. 115V 400 Hz - A 3-phase 15A toggle circuit breaker shall be provided for the 115/208 400 Hz power.

b. 28 VDC - A 10A toggle circuit breaker shall be provided for 28 VDC power.

c. Phase sequence relay - A phase sequence relay shall be provided to prevent power actuation in the event of a phase reversal or loss of any phase of the 3-phase 115V power.

d. UUT Power Contactor - A contactor under control of the UUT Power Switch shall be provided.

3.7.1.5.2 Reader, Punched Tape

3.7.1.5.2.1 Function

The photoelectric tape reader shall convert information stored as an array of punched holes on tape to electrical out-

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3.7.1.5.2.1 (Continued)

put signals which are used by other electrical equipment for purposes of computations or control. The reader shall be used in ground support equipment in accordance with military specifications and standards.

3.7.1.5.2.2 Form Factor

The reader shall not exceed 3.5 by 19 by 5.7 inches. The weight of the unit shall not exceed 25 pounds.

3.7.1.5.2.3 Characteristics

The reader shall exhibit the following characteristics for compatibility with the MCS Test Set Group.

a.	Power Input:	105-134 VAC, 50-420 Hz, single phase, 125W
b.	Start Time:	Less than 25 Milliseconds. With pinch roller engaged, less than 4 milliseconds.
c.	Stop Time:	Less than 200 Microseconds
d.	Stopping Characteristics:	On character to full rated speed, run command must be removed within 80 microseconds after sprocket leading edge.
e.	Sprocket Hole Rise Time:	Less than 5 Microseconds
f.	Output Control Logic:	"HOLE" (1) positive: +2.4 V Min. @-400 Microamperes "NO-HOLE" (0) positive: +0.4V Max. @ 16 Milliamps

g. The reader shall be a single line: bi-directional positive output, positive command photo reader.

h. Tape, Size & Thickness; 1" EIA standards, 8 level and sprocket, 50% opacity, .0025 - .004 thickness. No adjustment required for thickness or opacity.

i. Reading Speed:

1) Asynchronous (step): 0-300 CPS

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3.7.1.5.2.3 (Continued)

2)	Synchronous (slew):	625 ± 50 CPS at 60 Hz line frequency
3)	Rewind Speed:	62 inches per second (625 CPS)
4)	Tape Leader:	48" minimum
5)	Tape Trailer:	48" minimum

3.7.1.5.2.4 Controls

The following controls shall be provided on the front panel. Reader response to the controls shall be as listed in a, b, and c below:

a. Load/Run - A slide switch shall remove reader power in the LOAD position and apply reader power in the RUN position. When on, the reader capstans will rotate and the photoelectric light source will illuminate.

b. Forward direction switch - A pushbutton switch shall allow manual advance (left to right motion) of the tape.

c. Reverse direction switch - A pushbutton switch shall allow manual rewind (right to left motion) of the tape.

3.7.1.5.3 Transport, Tape

3.7.1.5.3.1 Function

The tape handler shall be used in conjunction with the photoelectric tape reader. It shall provide the means for storing, paying out, and taking up long lengths of punched tape which pass through the reader during the reading operation.

3.7.1.5.3.2 Form Factor

The transport shall not exceed 8.75 by 19 x 8.5 inches. The weight of the unit shall not exceed 30 pounds.

3.7.1.5.3.3 Characteristics

The transport shall exhibit the following characteristics for compatability with the MCS Test Set Group.

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3.7.1.5.3.3 (Continued)

a.	Power Input:	105/134 VRMS, 50/400 Hz, 125W
b.	Tape:	1" EIA standard (narrower tapes may be used)
c.	Tape Speed:	62" inches ± 5"/sec
d.	Tape Capacity:	1100 Feet (.0025) 700 Ft. (.004) NAB reel
e.	End of Tape/Broken Tape:	Power switch shut off by take up arm.
f.	Maximum Step Speed:	0-50-0 inches/sec. intermittently
g.	Reel Size:	8 inches

3.7.1.5.3.4 Controls

a. The Power switch shall turn off all power to the handler and/or the reader.

b. The rewind direction switch shall permit selection of fast rewind in either the forward or reverse direction.

c. The Mode switch shall have 3 functions:

1) Power on for both reader and handler (normal operating mode).

2) Power on for reader only (for loop tape operations).

3) High speed manual rewind (direction set by rewind direction switch).

3.7.1.5.4 Computer, Test

3.7.1.5.4.1 Function

The test computer shall consist of a control panel and a general purpose digital computer. The test computer shall include memory of 8,192 words of 16 bit length. The test computer shall contain the necessary electronics to perform the following functions:

a. Accept test control instructions from the test operator.

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3.7.1.5.4.1 (Continued)

b. Display test progress and results.

c. Self test upon command and provide a display of the self test status.

d. Accept digital inputs from other systems.

e. Output digital data to other systems.

3.7.1.5.4.2 Form Factor

This test computer shall not exceed 10.5 by 19 by 15 inches. The weight of the unit shall not exceed 50 pounds.

3.7.1.5.4.3 Characteristics

a. Computer performance - The test computer shall be a general purpose, stored program digital computer and shall employ a binary fixed point, two's complement number system. All internal operations shall be performed in word parallel format. Necessary input-output control and timing shall be provided in the computer.

b. Functional Characteristics

1) Data Word Format - The computer data word format shall be as shown in Figure 1. Negative numbers shall be represented in two's complement format with a one in the sign bit.

2) Instruction Word Format - The computer instruction word format shall be as shown in Figure 1. A 16 bit base register shall be available to provide the capability to reference any memory location at the programmer's option. Use of all addressing modes other than the indirect mode shall require only one computer memory cycle time. Each level of indirect addressing shall add no more than one memory cycle time to the overall execution time.

3) Instruction Repertoire - The computer instruction repertoire shall contain, as a minimum, the operations with the maximum execution times as listed in Table I. The execution time shall include all the time necessary for the instruction read, interpret, and operand read as well as the

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MEMORY REFERENCING INSTRUCTION

16 13	12 10	9	8 1
INSTR	REGISTER	X	ADDRESS FIELD

SHIFT INSTRUCTION

16 13	12 10	9 6	5 1
INSTR	REGISTER	N-1	SHIFT TYPE

SKIP INSTRUCTION

16 13	12 10	9	8 1
INSTR	REGISTER	X	SKIP CONDITION CODE

I/O INSTRUCTION

16 13	12 10	9	8 1
INSTR	REGISTER	N/A	SELECT CODE

BINARY DATA WORD

16 1

FRACTIONAL DATA WORD

16	15 1
S	FRACTION

FIGURE 1

INSTRUCTION WORD FORMAT

(AND DATA)

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CODE	MNEMONIC	INSTRUCTION	INDEXING	TIME*
00		Load HC		
01	XB	Exchange Block Register	N/A	2 usec
02	SH	Shift	N/A	2 usec
03	SK	Skip	N/A	2 usec
04	JC	Jump Back & Count	N/A	2 usec
05	J	Jump (Inside Block)		2 usec
06	I	Input		2 usec
07	O	Output		2 usec
10	LD	Load		3 usec
11	ST	Store		3 usec
12	A	Add		3 usec
13	S	Subtract		3 usec
14	M	Multiply		8 usec
15	D	Divide		17 usec
16	N	NAND		.3 usec
17		Store Hc		

*Includes indexing and 1 usec for the instruction fetch and address computation cycles.

TABLE I

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3.7.1.5.4.3 (Continued)

performance of the instruction itself. The computer shall be capable of performing the instruction repertoire in any sequence with any operand address or data word bit pattern without malfunction or erroneous result.

4) Interrupts - An interrupt structure shall be provided which shall cause the computer to be automatically interrupted and the program transferred to a unique memory location from which a user subroutine can be initiated to service the interrupt. The detection of acceptable power supply status or the actuation of the program LOAD switch shall cause the computer to enter its interrupt routine.

5) Timing - The computer shall contain a crystal oscillator for use in accurate time determination. A 50 microsecond interval derived from this source shall be used to generate the interval timer interrupt. The accuracy of the source frequency and derived timing signals shall not vary more than $\pm$ 100 parts per million over any one hour operating period for any of the conditions specified herein. Unless otherwise specified, all timing and control signals necessary to accomplish input/output data transfers shall be provided by the computer and derived from this timing source.

6) The memory for all units shall be an 8192 word randomly addressable subassembly. All words shall contain 16 bits per word, including sign. The truth table for a 512 word Read-Only-Memory shall be as required to perform tape load functions.

7) Input-Output - The test computer shall contain an input-output unit that shall provide the signal interface between the computer and all other devices. All computer inputs and outputs shall be set to zero at power turn-on and shall be enabled under program control. Externally mounted connectors shall provide a digital data bus (digibus) interface with all other devices.

8) Digibus Structure - The I/O bus structure consists of eleven line pairs. Eight are data

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3.7.1.5.4.3 (Continued)

lines (referred to as the Data Bus or Digibus) and three are control lines necessary for bus operation. Each time a peripheral is addressed by the computer a command code will be placed on the Digibus. This will be followed by a data word on the same bus as directed in Figure 2.

The data word will be placed on the bus by either the computer I/O or the addressed peripheral depending on whether the computer had requested an input from the peripheral or indicated an output to the peripheral during the previous command code. All timing controls and commands will originate with the computer. A peripheral cannot initiate bus command codes, but it may put data on the bus when instructed to do so. The command-code provides the peripheral three distinct bits of information. Namely, bits 4 thru 7 indicate the peripheral address, bits 1 thru 3 are used within the peripheral as sub-addresses or functions within the peripheral, bit 8 is not used. The three control lines to the peripheral are Command Strobe, Data Strobe and Input Enable.

(A) Command Strobe - The command strobe is centered within the time period that the command code is on the I/O bus. This strobe is sent to all peripherals indicating there is a command code on the bus. At the trailing edge of this pulse, each peripheral must decode the I/O bus to determine if the address on the bus corresponds to its assigned address. If the address corresponds, the peripheral must accept or send the following data word on the I/O bus; if the address does not correspond the peripheral will ignore the immediately following data word.

(B) Data Strobe - The data strobe is centered within the time period that the data word is on the I/O bus. If the peripheral address has been decoded and matches, the trailing edge of the data strobe is used to input and/or decode the data word.

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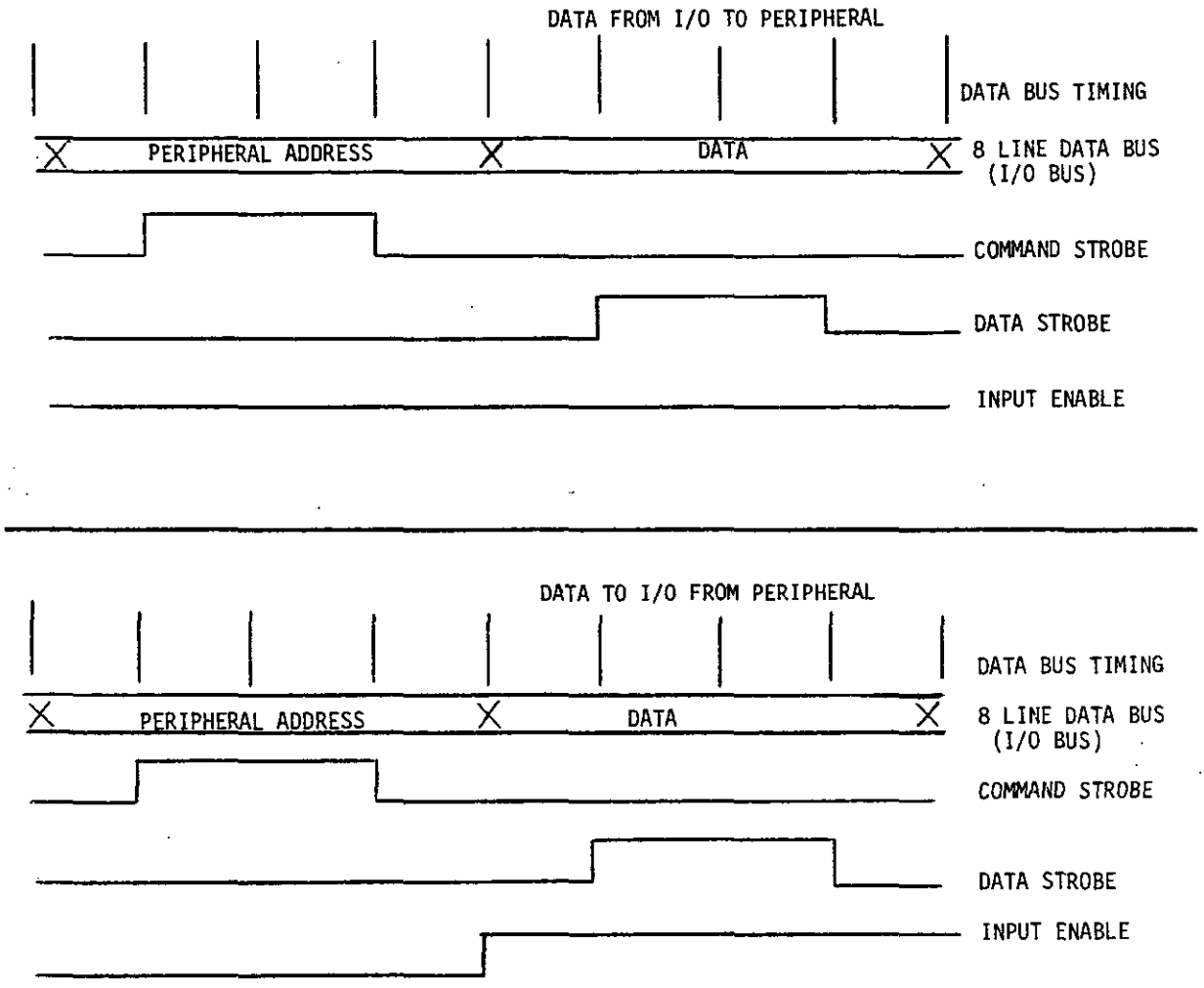


FIGURE 2

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3.7.1.5.4.3 (Continued)

(C) Input Enable Strobe - the input enable strobe is used to enable peripheral units to place data words on the bus. This strobe is up during the entire data word time period and is used to gate on the peripheral line transmitters (MD 7831's) when commanded by the computer. It is required that data to be sent to the computer be present for the full period that the strobe is enabled and that the data bits do not change during this period.

(D) Strobe Summary - The input enable strobe will only be present when the computer commands the peripheral to input a word into the I/O. The Command and Data Strobes will always occur when there is conversation on the I/O bus.

(E) Power Supply Performance - The test computer shall contain an integral power supply capable of accepting the power input specified in 3.6.8. The Power Supply (PS) shall convert this input power to the required DC regulated voltages necessary for test computer operation within specification requirements.

3.7.1.5.4.4 Controls

The following controls shall be provided on the front panel. Test computer response shall be as listed in the control descriptions below:

a. PROCEED - Momentary toggle switch used to start a test and to continue after either a programmed stop or operator stop.

b. STOP - ON/OFF switch. When ON, the program pauses at the end of the current test. When OFF, the program proceeds to the next test. If this switch is left ON, PROCEED can be used to single stop the program.

c. GO TO - ON/OFF switch used in conjunction with the 16 BIT octal data entry switch. When ON, the next overlay/test is determined by the content of the data entry switches. When OFF, normal program flow

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3.7.1.5.4.4 (Continued)

occurs. If STOP is OFF and GO TO is ON, the program will continually loop on the test selected by the data switches.

d. RESTART - Momentary toggle switch used to restart the complete test. This switch causes the tape to rewind, reload the test executive and wait for PROCEED.

e. DATA ENTRY - 16 bit octal lever type switch used to enter control information in the form of overlay and test number, or data information if the program requires it.

f. SENSE - ON/OFF switches perform functions which are determined by software as required. Two each are provided.

g. PROGRAM LOAD - Loads program from Tape Reader to PTP RAM.

3.7.1.5.4.5 Indicators and Displays

The following monitors, displays and incandescent indicating lights shall be provided on the front panel.

a. OVERLAY/Test Number - 6 digit octal display for overlay and test number.

b. TEST DATA - 6 digit octal display used to indicate test results, error codes, etc.

c. READY - Indicates program is ready to start testing or continue testing.*

d. TESTING - Indicates program is currently executing a test.

e. ENTER CONTROL - Indicates that GO TO is ON, STOP is ON, and the program is at the end of a test. The next test will be determined by the data switches.*

f. ENTER DATA - Indicates that a programmed pause has occurred and operator data is required via the data switches.*

g. GO - Indicates the last test completed without fault.

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3.7.1.5.4.5 (Continued)

h. FAULT - Indicates the last test completed with fault.

i. PAUSE - Indicates the test set is waiting for some operator action.*

j. PROCESSOR OK

k. LOADER FAULT

*If any of these indicators are lit, the proceed button must be pushed to start or continue with the test sequence.

3.7.1.5.5 Maintenance Panel

3.7.1.5.5.1 Function

The maintenance panel shall consist of a panel containing the controls, indicators and two logic circuit boards used to perform maintenance functions within the test set. The maintenance panel shall provide the capability to perform the following functions with the test computer (specified in 3.7.1.5.4).

a. Manual entry of instructions and/or data into registers or memory via the console keyboard.

b. Display of registers and memory contents.

c. Program execution in the normal mode (continuous) or manually-initiated single instruction mode (stepped) for software debugging.

d. Conditional halts and test points useful during program or processor checkout and maintenance.

3.7.1.5.5.2 Form Factor

The maintenance panel shall not exceed 3.5 x 11 x 4 inches. It shall be mounted on the front panel of the test computer, shall have a protective cover, which shall be closed when maintenance on the test set is not being performed. The weight of the maintenance panel shall not exceed 10 pounds. The two maintenance panel logic boards shall be compatible with the test computer card rack utilizing the same type and size connectors.

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3.7.1.5.5.3 Controls

The following controls shall be provided on the maintenance panel.

a. CLR - Momentary. CLR clears the processor instruction actuating counter (IC) to zero.

b. INIT - Momentary. If the HALT switch is off, depressing INIT starts program execution at the location specified by the instruction counter (IC). If the HALT switch is on, depressing INIT enables the single-instruction mode which halts program execution at the completion of each instruction.

c. Register Number - One rotary switch for each display. The position of these switches selects the register to be displayed. Each switch is analogous to bits 10, 11, and 12 of the instruction words. The rotary switch for display B also provides the lamp test function for the maintenance panel.

d. HALT COND A - Placing the HALT COND A switch in the active (up) position causes an automatic program halt when the contents of the keyboard register match the contents of the processor register selected by the Display A Register Number switches.

e. RTCEN (Real-Time Clock Enable) - In the off position, the real-time clock overflow signal is inhibited. Placing RTCEN in the active position enables the real-time clock overflow signal. If a real-time clock overflow (RTOF) occurs, program execution is halted and the indicator lamp above RTCEN is turned on to indicate that an RTOF has occurred and program execution halted.

f. HALT - Two-position switch. Placing the HALT switch in the up position halts program execution at the completion of the current instruction.

g. SENSE - Two-position switch. This switch is provided for the convenience of the programmer. Its position can be tested via the program.

h. Load/Display - Three momentary switches. These 3 switches are used during load and display register, memory, or block operations. These functions are:

1) THIS MEM-Activating THIS MEM upward loads the contents of the keyboard register into memory

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3.7.1.5.5.3 (Continued)

when memory position is selected on the REG A Rotary Switch. Depressing THIS MEM with MEM position selected on REG A Rotary Switch displays the memory location sepecified by the EA register.

2) NEXT MEM - Depressing NEXT MEM displays the memory location specified by the EA register plus 1.

3) REG A-Activating REG A loads the contents of the keyboard register into the register specified by the Display A Register switch. Depressing REG A displays the contents of the register selected by the REA A Rotary Switch.

i. Keyboard (not labeled) - 9 pushbuttons. The keyboard is used during load or examine memory, and conditional halt operations. The functions are:

1) CLEAR - Depressing CLEAR resets the keyboard register to 0 and causes the contents of the keyboard register to be displayed on Display A. It also enables the keyboard register such that the next digit buttons depressed are entered into the keyboard register.

2) 0 through 7 - Depressing any of these 8 (octal) digit buttons after depressing CLEAR causes entry of that digit into the keyboard register, starting with the most significant bit.

j. Maintenance Panel Interlock (Not Labeled) - Pushbutton. Closing of the maintenance panel protective cover activates this switch which causes the HLT FLAG to the test computer to be disabled and power to the maintenance panel to be disconnected.

3.7.1.5.5.4 Indicators and Displays

The following monitors, displays and LED indicating lights shall be provided on the maintenance panel.

a. DISPLAY A - A 6-digit light emitting diode (LED) display. The 6 octal digits displayed represent the 16-bit configuration of the register specified

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3.7.1.5.5.4 (Continued)

by the Display A Register switch, the block register, the memory register, or the keyboard register. (The most significant bit is the left most digit. Therefore, the left most digit can take on the values of 0 or 1 only.)

b. DISPLAY B - A 6 digit LED display. The 6 octal digits displayed represent the 16-bit configuration of the register specified by the Display B Register Number switch.

c. OP CODE - Two digit LED display. The 2 octal digits represent the 4-bit configuration of the current operation code.

d. REG (register) - One digit LED display. This octal digit indicates the register specified by the current instruction.

e. INDEX - LED Indicator Light. This light indicates whether or not indexing is specified by the current instruction.

f. RTOF - Indicator lamp. The RTOF lamp is turned on whenever a real-time clock overflow occurs.

g. CARRY - Indicator lamp. The CARRY lamp is turned on whenever a carry out of the 16th bit is generated by the processor arithmetic unit.

h. OVFL - Indicator lamp. The OVFL lamp is turned on whenever an overflow occurs during arithmetic operations.

i. SHIFT - Indicator lamp. The SHIFT lamp is turned on whenever the bit shifted out of a register is a 1.

j. CONDA - Indicator lamp. The CONDA is turned on whenever the HALT CONDA switch is active and a condition A occurs.

k. HLT FLAG - Indicator. The HLT Flag lamp is turned on whenever the HALT, RTOF, or HALT CONDA are active and the maintenance panel has issued a halt to the test computer.

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3.7.1.5.6 Converter, Interface Unit

3.7.1.5.6.1 Function

The Interface Unit Converter shall accept commands and controls from the test computer via the DIGIBUS. It shall simulate the normal analog/digital inputs to the AWG-10A System Computer and convert the UUT outputs into digital format for evaluation by the test computer. In testing the ability of the UUT to correctly interpret and react to signals from the radar system, the Interface Unit Converter shall, upon receiving instructions from the test computer, convert a digital signal to the required input signal and value and then direct this signal to the proper input pin of the UUT I/O section.

Analog input - The digital instruction is changed to analog signals by the Interface Unit Converter switching matrix which is also controlled by the test computer.

Digital input - The digital input data is received by the Interface Unit Converter in parallel bit format, stored in a shift register and is then transmitted to the UUT as serial data.

Miscellaneous input - The digital information from the test converter is appropriately converted to AC, DC, discrete or simulated synchro signals and directed to the input section of the UUT.

The Interface Unit Converter also receives output signals from the UUT, converts them to a parallel bit format and sends them via the DIGIBUS to the test computer for evaluation.

Analog outputs - Changed to digital data via the Interface Unit Converter analog/digital converters prior to transmission to the test computer.

Digital output - The serial output data is buffered and converted to parallel bit format and then transmitted to the test computer.

Miscellaneous output - Received by the Interface Unit Converter, and after being translated to digital data via electronic converters and digital logic elements are sent to the test computers.

3.7.1.5.6.2 Form Factor - The Interface Converter shall not exceed 11 by 20 by 19 inches. The weight of the unit shall not exceed 35 pounds.

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3.7.1.5.6.3 Interface Unit Converter to UUT Interface - The Interface Unit Converter shall provide processing of all active signal lines to the UUT connectors.

3.7.1.5.7 Blower Assembly

3.7.1.5.7.1 Function - The blower assembly shall supply cooling air to all units within the full bay of the MCS test set group.

3.7.1.5.7.2 Form Factor - This unit shall not exceed 7 x 18.5 x 19 inches. The weight of the unit shall not exceed 36 pounds.

3.7.1.5.7.3 Characteristics -

a. General. The unit shall be in accordance with MIL-T-28800, Class 3.

b. Electrical.

Voltage: 115 VAC
Phase: Single
Frequency: 400 Hz
Internal Start-Up Capacitor
Power Consumption: 650 Watts
Current: 6.8 AMPS

c. Mechanical.

Blower assembly shall be capable of being mounted in a standard 19 inch rack.

Motor Speed: 3700 RPM
Capacity: 550 CFM
Electrical connections shall be via no. 6 screws on terminal block.

3.7.1.5.8 Power Supply Assembly

3.7.1.5.8.1 Function - The Power Supply Assembly shall be a rack mounted unit for use in Ground Support Equipment in accordance with military specifications and standards.

3.7.1.5.8.2 Form Factor - The Power Supply Assembly shall not exceed 8.75 x 20 x 19 inches (not including slide rails). The weight of the assembly shall not exceed 38 pounds.

3.7.1.5.8.3 Characteristics -

a. General. The Power Supply Assembly shall be in accordance with MIL-T-28800, Class 3.

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3.7.1.5.8.3 (Continued)

b. Electrical.

1) Power Input

(A) Voltage and Phase: 115 volts RMS ± 10%, (line to neutral), three phases, all phases equally loaded except fan on one phase.

(B) Frequency: 380 to 420 Hz

(C) Harmonic Content: Not more than 5%, total root mean square (RMS).

2) *Power Output*

+12 VDC @ 1.0 AMP
+25 VDC @ 3.0 AMP } has common ground
-25 VDC @ 3.0 AMP }
+60 VDC @ 1.0 AMP } has common ground
-60 VDC @ 1.0 AMP }
+35 VDC @ 5.0 AMP
+ 5 VDC @ 20.0 AMP

3) Remote sensing capability for each output.

4) Regulation: line and load

+12 VDC : 0.1%
+25 VDC : 0.01%
-25 VDC : 0.01%
+60 VDC : 0.01%
-60 VDC : 0.01%
+35 VDC : 0.1%
+ 5 VDC : 0.1%

5) Ripple: 0.1% or 50 millivolts max.

6) Response Time: less than 20 microseconds

7) Efficiency: 50% or greater

8) Temperature Coefficient: 0.01%/°C

9) Stability: 0.1%

10) Adjustment: -5% to + overvoltage

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3.7.1.5.8.3 (Continued)

11) Overcurrent Protection. The units shall not be damaged due to either a short duration or an indefinite short circuit, or by continuous overload. The output current shall limit to a value not to exceed 150% of rated current (per 2 above).

12) Overvoltage Protection. Under all conditions of load, environmental and failure or malfunction of parts, the outputs shall not exceed the absolute maximum values listed for a period greater than 10 microseconds.

13)

Output	Absolute Maximum Voltage
+12 VDC	+14 VDC
+25 VDC	+27.5 VDC
-25 VDC	-27.5 VDC
+60 VDC	+68 VDC
-60 VDC	-68 VDC
+35 VDC	+40 VDC
+ 5 VDC	+ 7 VDC

c. Mechanical

1) The power supply shall be designed for front panel mounting (standard 19 inch) and for slide mounting. There shall be two handles on the front panel.

2) The front panel shall be designed such that when installed in a console with proper gasketing, the power supply assembly shall meet the electromagnetic interference requirements of MIL-STD-461.

3) Test points for all outputs shall be easily accessible when the power supply assembly is extended on slides.

3.7.1.5.9 Connector Plate Assembly

3.7.1.5.9.1 Function - The connector plate assembly provides the junction point between inter-console wiring and extra-console wiring of the MCS test set group.

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3.7.1.5.9.2 Form Factor - This unit shall be an aluminum plate whose dimensions shall not exceed 8.5" by 4.5" by .125" thickness. Its weight shall not exceed 15 pounds.

3.7.1.5.9.3 Characteristics - The following cables shall be joined at the connector plate assembly. The bulkhead connector attached to this unit shall be on the inter-console cable.

Inter Console Cable Connector		Extra Console Cable Connector	
Cable	Connector	Cable	Connector
W1	A8J1	W107	A8P1
W4	A8J2	W102	A8P2
W5	A8J3	W105	A8P3
W6	A8J4	W104	A8P4
W4	A8J5	W108	A8P5
W9	A9J6	W103	A8P6
W8	A8J7	W106	A8P7
W7	A9J8	-	-

3.7.1.5.10 Fixture Cooler, Air

3.7.1.5.10.1 Function

The air fixture cooler shall provide mounting for the Interconnecting Box and LRU 15/16 during test. Self contained in the fixture shall be a motor and fan assembly to provide cooling air to the unit-under-test and interconnecting box. When mounted in the fixture, the LRU's shall be positioned such that the MCS test set group cables can be used to interconnect LRU's 15 and 16.

3.7.1.5.10.2 Form Factor

The air fixture cooler shall not exceed 27 x 25 x 4 inches. This unit weight shall not exceed 40 pounds.

3.7.1.5.10.3 Characteristics - The fixture shall exhibit the following characteristics:

a. A cable trough shall be provided to protect the extra console cables.

b. Guide pins and holddowns to mechanically align the UUT with the fixtures air outlets shall be provided.

c. The fixture's fan shall provide a minimum of 1.8 pounds per minute of cooling air.

d. The fixture shall provide mounting provisions for the Interconnecting Box such that it may be interconnected with the UUT by a cable whose maximum length is 12 inches.

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TABLE 2

INTERCONNECTING BOX

SUBCOMMANDS

COMMAND NAME	COMMAND BINARY EQUIVALENT BIT 3	BIT 2	BIT 1	FUNCTION INTERCONNECTING BOX INPUT	FUNCTION INTERCONNECTING BOX OUTPUT
C0	0	0	0	Write CON1-8	Read CON1-8
C1	0	0	1	Write CON9-16	Read Status WD
C2	0	1	0	Write UUT Control Discretes	
C3	0	1	1	Write UUT Instruction Discretes	
C4	1	0	0	Write Interconnecting Box Control Reg	Read PA1-8
C5	1	0	1		Read PA9-16
C6	1	1	0		Read UUT Control BITS
C7	1	1	1		Read UUT Fault BITS

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3.7.1.5.11 Interconnecting Box

3.7.1.5.11.1 Function - The Interconnecting Box shall consist of the necessary circuitry to perform the following functions.

a. Interface the MCS test set group digibus.

b. Interpret and execute digibus commands.

c. Signal condition digibus data for transmission to the UUT.

d. Signal condition UUT data for transmission to the digibus.

e. Generate required timing and control signals for all data transmissions.

3.7.1.5.11.2 Form Factor - The Interconnecting Box shall not exceed 11 x 9 x 3 inches. The weight of the unit shall not exceed 6 pounds.

3.7.1.5.11.3 Characteristics

a. General - The Interconnecting Box shall be a digital buffer between the MCS test set group and LRU 15 test connector, J2. It shall be controlled by and transfer data over the tester digibus.

b. Control - The MCS test set group (hereinafter called TSG) shall control the Interconnecting Box via the TSG digibus. The following paragraphs will detail the bit assignments within the command word of the digibus transmissions.

c. Interconnecting Box Address - The Interconnecting Box digibus address shall be 03 (0011 for bits 7, 6, 5, 4, of the command word respectively).

d. Interconnecting Box Subcommand Assignments - Bits 1, 2 and 3 of the digibus command word shall be decoded to represent the functions listed in Table 2.

e. TSG Data - Data word transmissions between the TSG and Interconnecting Box allow the following functions to be performed.

1) Loading and monitoring of UUT registers.

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3.7.1.5.11.3 (Continued)

2) Loading and monitoring of UUT memory.

3) Single stepping of the UUT.

4) Application of UUT mode control signals.

5) Monitoring of UUT status signals.

6) Self test of the Interconnecting Box.

7) Monitoring of Interconnecting Box status.

f. *Timing - The Interconnecting Box shall develop no* unique timing internally. All timing shall be derived from either the test set or UUT interface signals.

g. Self Test - The Interconnecting Box shall contain the necessary circuitry to allow a TSG controlled test of the Interconnecting Box. Initiation of this test shall be in conjunction with all normal digibus signals to indicate proper operation of the Interconnecting Box on a GO/NO-GO basis to a 90% confidence level.

3.7.1.5.12 Interlock Assembly

3.7.1.5.12.1 Function - The interlock assembly shall provide the switch closure necessary to complete the power interlock system of the MCS test set group. This interlock shall be closed by either the shutting of the rear door of the test set group or by intentional override of the interlock when the rear door is open.

3.7.1.5.12.2 Form Factor

The form factor of the switch shall be per drawing 90C99015.

3.7.1.5.13 Top Plate Assembly

3.7.1.5.13.1 Function - The top plate assembly shall provide the prime power interface and filtering for the MCS test set group.

3.7.1.5.13.2 Form Factor - The top plate assembly shall not exceed 19 by 24 by 4 inches. The unit weight shall not exceed 8 pounds.

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3.7.1.5.13.3 Characteristics - The filter characteristics of the top plate assembly shall be as follows:

a. Filters shall meet the requirements of MIL-F-15733/2.

b. Electrical

Maximum current	20.0 Amperes
Rated Voltage	120 Volts AC 400 Hz 400 Volts DC
Inductance	22 UH
Series Resistance	.008 OHMS
Capacitance to Ground	2.8 UF

Min. Insertion Loss at 25°C (DB)	Frequency (MHz)
55	.15
90	.5
90	5
90	20
90	1000

3.7.1.5.14 Isolation Transformer

3.7.1.5.14.1 Function - The Isolation Transformer shall isolate Phase A return used by LRU 15 from test set chassis ground.

3.7.1.5.14.2 Form Factor - The Isolation Transformer shall not exceed 5.25 inches by 3.94 inches by 3.38 inches and shall contain four mounting studs, .5 ± .06 inches in length. Its weight shall not exceed 9 pounds.

3.7.1.5.14.3 Characteristics - The Isolation Transformer shall exhibit the following characteristics.

a. Electrical

Voltage, Primary	105/115/125	Volts RMS
Voltage, Secondary Within 5% tolerance when measured at full load and 115 VRMS input	115 CT	Volts RMS
Frequency, Operating	380-420	Hz
Current	4.4	VA
Percent Regulation, No Load to full load	10	Percent
Power	500	VA

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3.7.1.5.14.3 (Continued)

Dielectric Withstanding Voltage Winding to winding and winding case	1000	Volts RMS

b. Mechanical

Construction per MIL-T-27D, Grade 4, Class R

3.7.1.5.15 Cabinet, Electrical Equipment

3.7.1.5.15.1 Function - The electrical equipment cabinet shall provide the enclosure and mounting of all items specified in 3.7.1.5.1 thru 3.7.1.5.14. It shall also provide a slide out work surface.

3.7.1.5.15.2 Form Factor - The electrical equipment cabinet shall be a 1-1/2 bay console. Bay 1 shall not exceed 73 x 23 x 26 inches and Bay 2 shall not exceed 34 x 27.5 x 26 inches. The unit weight shall not exceed 600 pounds.

3.7.2 Test Set, Optical Sight, AN-AWM-81 - The Test Set shall meet the following requirements.

3.7.2.1 Function

The Test Set, Optical Sight, shall be utilized in the intermediate level shop for testing and troubleshooting the AN/AWG-10A Servoed Optical Sight, LRU 9A5. The test set shall receive electrical power from the MCS Test Set Group, OQ-230/AWM-80 or from standard navy bench outlets which provide 3 phase*, 400 Hz and 28 VDC shop power. The test set shall, in turn, provide the necessary stimuli and mounting facilities for operation of the optical sight assembly and shall either directly monitor or provide test points for monitoring of the outputs of this assembly.

3.7.2.2 Form Factor

The test set shall consists of two units:

3.7.2.2.1 Instrument Assembly, TS-3496/AWM-81, shall be contained in a case whose maximum dimensions shall not exceed 12.7 inches in width by 12.4 inches in height by 8.1 inches in depth. Handles on the front shall provide protection of the front panel controls and shall facilitiate lifting and transporting of the assembly within the shop area. When not in use, the instrument assembly shall be stored in a drawer of the AN/AWM-80 Test Set cabinet.

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* Note: Only one phase of the 3 phase power source is required for operation of this test set.

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3.7.2.2.2 Holding Fixture, MT-4817/AWM-81, shall consist of a movable plate, a vertical post and four horizontal extendable feet for stability. In its fully extended position the holding fixture shall not exceed nine inches in height and shall fit within a circle whose diameter shall not exceed 18 inches. When not in use, it shall be stored in a drawer of the MCS Test Set, AN-AWM-80.

3.7.2.3 Weight - The weight of the test set shall not exceed the following values:

Instrument Assembly, including cables - 30 lbs.
Holding Fixture - 30 lbs.

3.7.2.4 Content - The test shall contain the following circuits:

3.7.2.4.1 AC Power Supply. Shall supply 28 $^{+1}_{-1.8}$ VAC, 400 Hz single phase at a maximum of 8 amperes.

3.7.2.4.2 DC Power Supply. Shall provide two dc voltages of positive and negative 12 ± 0.12 VDC at a maximum of 100 milliamperes.

3.7.2.4.3 Annunciator Lamp Test Circuit. Shall consist of a voltage source of 5V, 400 Hz at a maximum of 500 milliamperes and a rotary selector switch to selectively apply the 5V excitation to individual lamps in the optical sight under test. The rotary switch shall have 9 positions plus an OFF position.

3.7.2.4.4 Mode Switch logic monitor. Shall consist of three lamps and a logic chart to indicate which lamps should be illuminated for specified positions of the mode switch on the optical sight assembly.

3.7.2.4.5 Reticle and Range Bar control circuit. Shall consist of three center tapped potentiometers for control of the reticle azimuth and elevation and the range bar displays on the optical sight assembly.

3.7.2.4.6 Potentiometer Test Circuit. Shall provide easily accessible test points and a four position (plus an OFF position) rotary selector switch for testing the indicator control potentiometers on the optical sight assembly.

3.7.2.4.7 CUE Lamp and Computer Operate Command functions. Shall provide test capability of these functions of the optical sight by simulating presence or absence of these command signals.

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3.7.2.4.8 Operate/Erase Switch Test. Two lamps shall monitor the functions of the Operate/Erase Switch on the optical sight assembly.

3.7.2.5 Controls - The Test Set shall contain the following controls.

3.7.2.5.1 POWER - ON-OFF Switch for power to the test set and to the unit-under-test.

3.7.2.5.2 LAMP TEST - 10 position rotary switch to sequentially energize the annunicator lamps on the UUT.

3.7.2.5.3 OPERATE - SPST switch to simulate loss of computer control of the optical sight.

3.7.2.5.4 CUE LAMP - SPST switch simulating weapon release command from the computer to the optical sight.

3.7.2.5.5 RTCL AZ Potentiometer Control simulating azimuth command to the optical sight reticle.

3.7.2.5.6 RTCL EL - Potentiometer control simulating elevation command to the optical sight reticle.

3.7.2.5.7 RNG ANAL BAR - Potentiometer control simulating command voltage to the Range Analog Bar of the optical sight.

3.7.2.5.8 POTENTIOMETER TEST - 5 position rotary switch to select a potentiometer in the optical sight and provide access for testing the operation of that potentiometer.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

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4.1.1 (Continued)

First article (preproduction) tests
Initial production tests
Acceptance tests
Life tests

4.2 First article (preproduction) tests - First article (preproduction) tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 First article (preproduction) test data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of tests - The tests shall include all tests deemed necessary, by the procuring activity, to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Within the limits specified in MIL-T-28800, first article tests shall include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

4.2.2.1 Maintainability assurance test - A maintainability assurance test shall be conducted in accordance with MIL-STD-471 to demonstrate the requirements of 3.4.2. The test details shall include as a minimum, task selection, test method, demonstration support material, demonstration conditions, demonstration test team, test schedule and test reports, and shall be a part of the test procedure, in accordance with 4.6, to be submitted to the procuring activity for approval.

4.2.2.2 Probability of successful performance test - A probability of successful performance assurance test to demonstrate the requirements of 3.4.3 shall be conducted. The test procedure shall be prepared in accordance with 4.6 and submitted by the contractor to the procuring activity for approval.

4.2.3 First article (preproduction) approval - Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the

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4.2.3 (Continued)

approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The test sample shall not be considered as one of the equipments under the contract.

4.2.4 Production equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved test sample. Each equipment shall be capable of successfully passing the same tests as imposed on the test sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production tests - One of the first ten production equipment shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The test sample shall not be selected for this test.

4.3.1 Scope of tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial production sample approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of initial production test sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.4 Acceptance tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commerical testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

Individual tests	(4.4.1)
Sampling tests	(4.4.2)
Reliability assurance test	(4.4.3)
Special tests	(4.4.4)

4.4.1 Individual tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

Examination of product	(4.4.1.1)
Operational test	(4.4.1.2)
Manufacturing run in test	(4.4.1.3)

4.4.1.1 Examination of product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

Note: Approval of the operational test procedure must be obtained from the Naval Air Systems Command prior to testing.

4.4.1.3 Manufacturing run in test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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4.4.1.3 (Continued)

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operating within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over; except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial production test (4.3) is deleted, then a Sample test shall be conducted on one equipment from the first 10 produced.

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4.4.2.1 Scope of tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests shall be conducted as required by both the Qualification Phase and Sampling Phase.

4.4.3.1 Qualification phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipment to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level A shall be used. The Accept-Reject criteria for Test Plan III shall be used.

4.4.3.2 Reliability sampling phase tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase Test Level A shall be used. The Accept-Reject criteria for Test Plan IV shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot size for sampling phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The sample shall be tested until an accept or reject decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly to the

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4.4.3.2.1 (Continued)

procuring activity, and to the Naval Air System Command, Attention: AIR-5344. The procuring activity reserves the right to stop the acceptance of equipment any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control etc., so that the entire contract quantity will show an accept decision.

4.4.3.3 Test details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicated additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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4.4.5 (Continued)

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in failure tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following (equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests). When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipment equal to, or more than, that listed below received at last 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient
Altitude	Normal ground (0-5000 ft.)
Humidity	Room Ambient
AC Voltage	115 ± 5 volts (380 Hz)
DC Voltage	27.0 ± 2.0 volts
Vibration	None

4.5.2 Test periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of life test samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test procedures - The procedures used for conducting preproduction test, acceptance test and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test

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4.6 (Continued)

procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of test equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5.0 PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

5.2 Marking - In addition to any special marking by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6.0 NOTES

6.1 Intended use - The AN/AWM-80 Missile Control System Test Set is to be used at the intermediate level maintenance shop, both in land based and shipboard applications. It shall be specified in accordance with this usage requirement as MIL-T-28800 Class 3 ground support equipment. The test set shall be used to test and fault isolate failures within the

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6.1 (Continued)

computer assemblies (LRU's 15 and 16), the Control Indicator, Test (LRU 14) and the Servoed Optical Sight (LRU 9A5) of the AN/AWG-10A Missile Control System. It provides the capability of exercising these units independent of the remainder of the assemblies of the AWG-10A System.

6.2 Performance objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-28800, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable subassemblies - As a general rule, nonrepairable subassemblies are encapsulated or hermetically-sealed. The number of connections internal to the subassembly have been held to a minimum. Detail parts tolerances and ratings have been so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions the nonrepairable subassemblies evidence a mean-time-to-failure greater than 5,000 hours, and for many applications this figure is nearer 50,000 hours.

6.4 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Ordering data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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6.5 (Continued)

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1).

6.6 Type designations - The parentheses ( ), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.5. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated equipment - The equipment shall operate with the following associated equipment (or equivalent).

Oscilloscope, OS-180A/AWM-21
Voltmeter, Digital, ME-371/AWM-21
Multimeter, AN/PSM-4B

6.9 Specification cognizance. This specification is under the cognizance of AIR-5344.

Preparing Activity:

NAVY - AS
Project No. 4935-N001

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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